Faculty of Computing

University of Sri Jayewardenepura

Undergraduate Final Year Project -Proposal Form

Please complete all of the following sections.			
Candidate Details			
Student Full Name:		Karunarathnage Uththara Maheshi Karunarathna	
Registration No.		FC110546	
Examination No.		FC211019	
Supervisor Details			
Supervisor (1) Name	Dr.Nadeeka De Silva	Department	Department of Mathematics, Faculty of Applied Sciences
Supervisor (2) Name	Ms. Dilmi Praveena Kulugammana	Department	Department of Knowledge Engineering and Communication
Project Details			
Title	Neuro-Symbolic Goal-Driven Restaurant Conversational Agent using Retrieval-Augmented Generation, Large Language Models, and Answer Set Programming		
Objectives	1. To design, develop, and evaluate, within the project duration, a Neuro-Symbolic restaurant conversational agent that integrates Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Answer Set Programming (ASP), achieving accurate and goal-driven restaurant-related conversations with a measurable task completion rate of at least 85%. 2. To implement and evaluate a Retrieval-Augmented Generation mechanism capable of retrieving restaurant menus, ingredients, nutritional information, and restaurant policies, improving response relevance by at least 80% and reducing hallucination rate by at least 30% compared to a standalone LLM baseline during conversational interactions. 3. To integrate and evaluate an Answer Set Programming module into the conversational agent to validate food recommendations against predefined constraints such as budget limits, dietary preferences, allergy restrictions, and restaurant policy rules, achieving a constraint satisfaction rate of at least 90% and ensuring logically consistent outputs across all test scenarios.		

	4. To evaluate the performance of the proposed Neuro-Symbolic conversational agent against a standalone LLM-based system using measurable evaluation metrics including accuracy, hallucination rate, constraint satisfaction, task completion rate, response relevance, and user satisfaction, demonstrating a statistically significant improvement of at least 20% across the majority of metrics. 5. To design and evaluate an explainability mechanism for the conversational agent that generates logical reasoning traces and validated recommendations, enabling users to understand how decisions are produced by the system, and achieving a user-reported explainability satisfaction score of at least 75% in post-interaction evaluation.
Project proposal	Refer Appendix II for project proposal guidelines. Please attach the Project Proposal to this form in the time of submission.
Second Section (Only accessed by the Supervisors)	
Filled by Supervisor	
Decision	• Approved without changes • Approved with minor or major revisions • Rejected, requiring a new submission
Comment	



Neuro-Symbolic Goal-Driven Restaurant Conversational Agent using Retrieval-Augmented Generation, Large Language Models, and Answer Set Programming

Faculty of Computing
University of Sri Jayewardenepura

Final Year Project Proposal

Name: K.U.Maheshi Karunarathna
Registration No.:FC110546
Examination No.: FC211019

Year:2026

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1. Introduction

Artificial Intelligence (AI) has experienced significant advancements with the development of Large Language Models (LLMs), which demonstrate powerful capabilities in natural language understanding, text generation, and conversational interactions [1]. Modern LLMs such as GPT and Llama models have shown remarkable performance in human-like communication and semantic understanding [2]. Due to these capabilities, LLM-based conversational agents are increasingly used in domains such as customer service, healthcare, education, and recommendation systems [7].

Despite their strengths, standalone LLM systems face several limitations. Since they operate as probabilistic models, they often generate hallucinated responses, produce inconsistent reasoning, fail to strictly satisfy user constraints, and provide limited explainability [3]. These challenges become more critical in goal-driven conversational environments where accuracy, logical consistency, and constraint adherence are important [6].

Answer Set Programming (ASP) is a symbolic reasoning approach capable of representing knowledge through logical rules and constraints [4]. ASP provides exact reasoning, explainable outputs, and strict constraint handling [5]. However, ASP systems have limitations when processing natural language and managing large volumes of unstructured information [9].

Retrieval-Augmented Generation (RAG) has emerged as an approach that improves AI systems by retrieving external knowledge during response generation [16]. RAG reduces dependency on internal model memory and improves factual grounding by accessing domain-specific knowledge sources [13]. Recent developments in Neuro-Symbolic AI attempt to integrate neural approaches and symbolic reasoning systems to combine linguistic flexibility with logical consistency.

This project proposes a Neuro-Symbolic Restaurant Conversational Agent integrating LLMs, RAG, and ASP to provide reliable restaurant recommendations while reducing hallucinations and improving explainability [11, 10]. The system will retrieve menu information and restaurant policies using RAG, interpret user intentions through LLMs, and validate recommendations using ASP reasoning [15, 21].

2. Problem Statement / Research Question

Current restaurant conversational systems mainly depend on probabilistic language models and encounter several limitations including:

- Generation of hallucinated menu recommendations
- Failure to satisfy dietary and budget constraints
- Lack of explainability in recommendation decisions
- Inconsistent reasoning during multi-step conversations

- Weak handling of restaurant policies and restrictions

These issues reduce the reliability and trustworthiness of restaurant recommendation systems.

Research Question

How can Retrieval-Augmented Generation, Large Language Models, and Answer Set Programming be integrated to improve reliability, logical consistency, and explainability in goal-driven restaurant conversational agents?

3. Objectives of the Project (Outcome-Based Objectives)

Objective 1

To design, develop, and evaluate, within the project duration, a Neuro-Symbolic restaurant conversational agent that integrates Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Answer Set Programming (ASP), achieving accurate and goal-driven restaurant-related conversations with a measurable task completion rate of at least 85%.

Objective 2

To implement and evaluate a Retrieval-Augmented Generation mechanism capable of retrieving restaurant menus, ingredients, nutritional information, and restaurant policies, improving response relevance by at least 80% and reducing hallucination rate by at least 30% compared to a standalone LLM baseline during conversational interactions.

Objective 3

To integrate and evaluate an Answer Set Programming module into the conversational agent to validate food recommendations against predefined constraints such as budget limits, dietary preferences, allergy restrictions, and restaurant policy rules, achieving a constraint satisfaction rate of at least 90% and ensuring logically consistent outputs across all test scenarios.

Objective 4

To evaluate the performance of the proposed Neuro-Symbolic conversational agent against a standalone LLM-based system using measurable evaluation metrics including accuracy, hallucination rate, constraint satisfaction, task completion rate, response relevance, and user satisfaction, demonstrating a statistically significant improvement of at least 20% across the majority of metrics.

Objective 5
To design and evaluate an explainability mechanism for the conversational agent that generates logical reasoning traces and validated recommendations, enabling users to understand how decisions are produced by the system, and achieving a user-reported explainability satisfaction score of at least 75% in post-interaction evaluation.

4. Brief Literature Background

Large Language Models have demonstrated strong capabilities in natural language processing and conversational interactions [2]. However, existing research highlights several weaknesses of LLM systems including hallucinations, weak logical reasoning, and lack of formal verification mechanisms [3].

Neuro-Symbolic AI has emerged as a promising approach that combines neural models and symbolic reasoning systems [7, 8]. Research studies such as VERUS-LM [11], Logic-LM [13], LLM-ARC [15], and related works [6, 17] demonstrate that integrating symbolic reasoning with language models improves logical consistency and reduces hallucinations.

Answer Set Programming has been widely used in knowledge representation and reasoning tasks due to its capability for exact logical reasoning and constraint satisfaction [4, 5]. ASP has shown effectiveness in planning, recommendation systems, and conversational state tracking [10, 9, 14]. Recent benchmarking studies further highlight both the potential and limitations of LLMs in solving ASP problems [20], while fine-tuning approaches such as LLASP [19] explore improving LLM performance on ASP tasks.

Retrieval-Augmented Generation improves AI systems by retrieving external knowledge during inference [16]. RAG increases factual accuracy and enables systems to utilise updated domain knowledge without retraining language models [21]. Ontology-based and neuro-symbolic integration approaches further enhance reasoning transparency and reliability in such systems [18, 16].

Existing studies mainly focus on general reasoning tasks, automated planning, and logic verification [3, 14, 15]. However, limited research exists on applying RAG-enhanced Neuro-Symbolic architectures to goal-driven restaurant conversational systems [10]. Therefore, this project aims to address this research gap by integrating LLMs, RAG, and ASP within a restaurant recommendation environment [11, 13].

5. Method/s

5.1 Research Methodology

This research adopts a **design and development methodology** combined with **experimental evaluation** to develop and assess the proposed Neuro-Symbolic restaurant conversational system.

The project development process consists of the following phases:

1. Dataset preparation and preprocessing
2. Retrieval layer implementation
3. LLM integration
4. ASP reasoning integration
5. Evaluation and performance analysis

5.2 Data Sources

The proposed system will utilize both structured and unstructured restaurant-related datasets collected from multiple sources to support recommendation generation and logical reasoning.

5.2.1 Structured Data

Structured data such as restaurant menu information, ingredient datasets, and nutritional information will be collected from real restaurant sources, publicly available food datasets, and online restaurant platforms. These datasets provide organized and machine-readable information required for recommendation and reasoning tasks.

5.2.2 Unstructured Data

Unstructured data including restaurant policies, discount policies, allergy guidelines, and user query datasets will be manually created and curated to simulate realistic conversational scenarios and reasoning conditions for the proposed system.

5.3 Tools and Technologies

The proposed system will utilize the following technologies:

Component	Technology
Frontend	Streamlit
Backend	Python
Large Language Model	Llama 3
Retrieval Framework	LangChain
Vector Database	ChromaDB
Embedding Model	Sentence Transformers
Reasoning Engine	Clingo (ASP Solver)
Data Processing	Pandas

Table 1: Tools and Technologies

5.4 Experimental Setup

Two systems will be compared:

5.4.1 Baseline System

LLM-only restaurant conversational agent.

5.4.2 Proposed System

RAG + LLM + ASP Neuro-Symbolic restaurant conversational agent.

The experiments will evaluate recommendation quality using restaurant scenarios involving:

- Budget restrictions
- Dietary preferences
- Allergy constraints
- Policy compliance
- Goal-driven conversations

5.5 Evaluation Metrics

The proposed system will be evaluated using:

- Hallucination Rate
- Recommendation Accuracy
- Constraint Satisfaction Accuracy
- Goal Completion Rate
- Response Time
- Explainability Performance

A comparative analysis of the solutions obtained by independent LLM systems, the proposed neuro-coding approach, and manual testing is performed.

The LLM component extracts user intentions and converts conversational inputs into structured constraints and symbolic representations. Retrieved restaurant knowledge and extracted constraints are then processed by the ASP reasoning engine to generate logically valid recommendations. Finally, the ASP outputs are transformed into natural language responses using the LLM, providing users with clear and relevant suggestions based on their preferences.

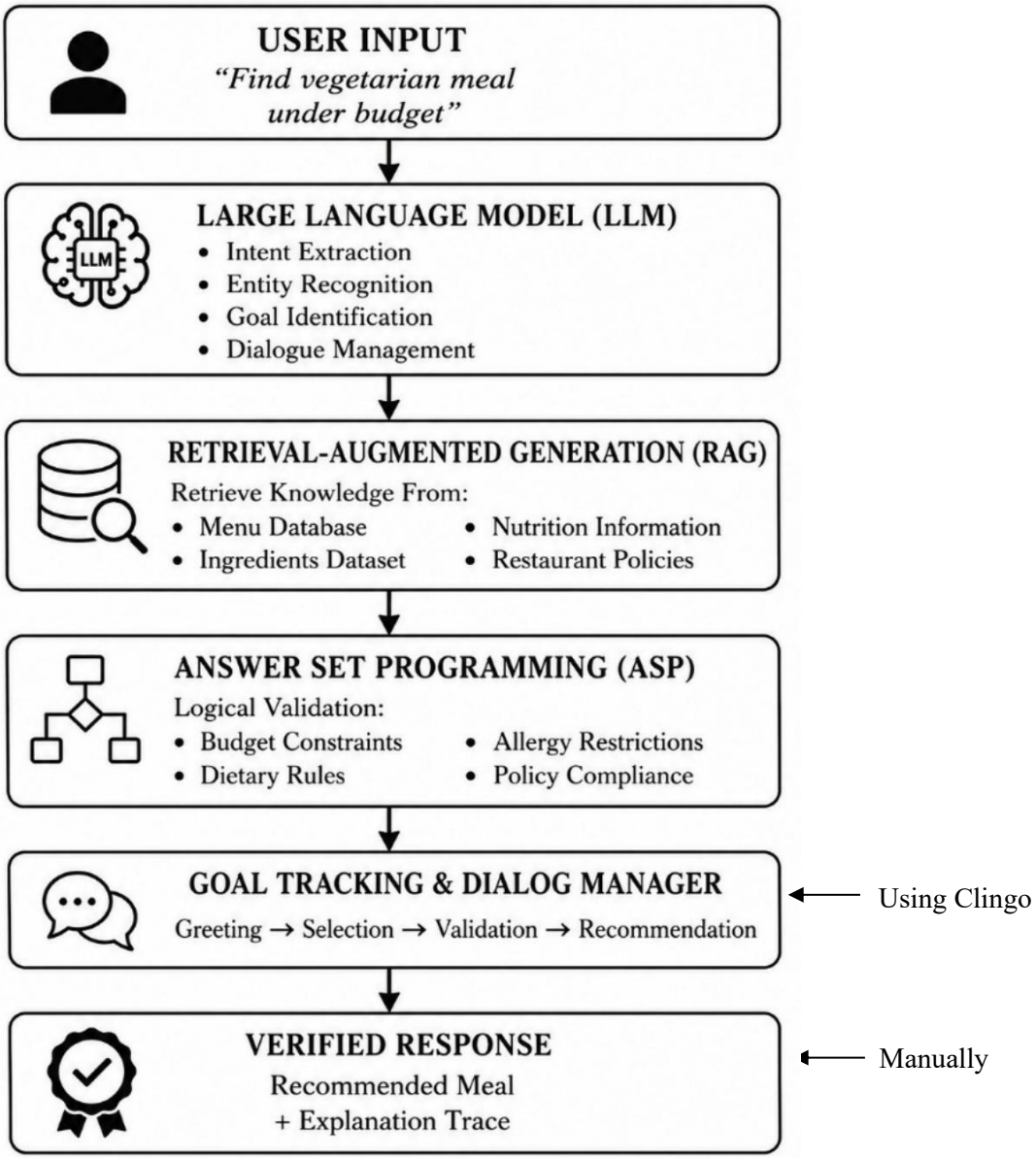


Figure 1: Proposed Architecture of Neuro-Symbolic Goal-Driven Restaurant Conversational Agent

6. Significance of the Research

This research is significant because it addresses major limitations of current AI-based restaurant conversational systems, including hallucinated responses, weak logical reasoning, poor constraint handling, and lack of explainability. Existing Large Language Model (LLM)-based systems often fail to consistently satisfy user requirements such as dietary restrictions, allergies, budgets, and restaurant policies, reducing their reliability in real-world applications.

The proposed Neuro-Symbolic conversational agent integrates Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), and Answer Set Programming (ASP) to improve recommendation accuracy, logical consistency, and transparency. The integration of

symbolic reasoning enables the system to validate recommendations against predefined constraints while reducing hallucinations.

In addition, the research contributes to the field of Neuro-Symbolic Artificial Intelligence by demonstrating a practical implementation of trustworthy and explainable AI in a goal-driven conversational environment. The findings may support future intelligent recommendation systems and other real-world AI applications requiring both natural language understanding and logical reasoning.

7. Timeline for Completion

Table 2: Gantt Chart (Project Plan)

No	Task	Duration (Weeks)	Start Date	End Date
Phase 1: Planning & Analysis				
1	Literature Review	4	May 2026	Jun 2026
2	Problem Analysis	2	Jun 2026	Jun 2026
3	Requirement Analysis	2	Jun 2026	Jul 2026
4	System Design	2	Jul 2026	Jul 2026
Phase 2: Development				
5	Dataset Collection	3	Jul 2026	Aug 2026
6	Data Preprocessing	2	Aug 2026	Aug 2026
7	RAG Module Implementation	3	Aug 2026	Sep 2026
8	ASP Module Implementation	3	Sep 2026	Sep 2026
9	LLM Integration	3	Sep 2026	Oct 2026
10	System Integration	2	Oct 2026	Oct 2026
11	Testing & Debugging	3	Oct 2026	Nov 2026
Phase 3: Evaluation				
12	Experimental Setup	2	Nov 2026	Nov 2026
13	Performance Evaluation	2	Nov 2026	Dec 2026
14	Comparative Analysis	2	Dec 2026	Dec 2026
15	Result Interpretation	1	Dec 2026	Jan 2027
Phase 4: Documentation				
16	Thesis Writing	4	Jan 2027	Feb 2027
17	Report Writing	4	Jan 2027	Feb 2027
Total Duration (Weeks)		**40**		

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9. Certification Section

I hereby certify to the best of my knowledge that the details mentioned above are true and accurate.	
Signature of Candidate	**Date**
	2026/05/25
I/ We hereby certify to the best of my/ our knowledge that the details mentioned above are true and accurate.	
Signature of Supervisor (s)	**Date**
	29/05/2026
Supervisor 2 	2026/05/28